# Tianhao Zhou (Ethan)

tianhao.zhou219@gmail.com | LinkedIn® Profile | 04 0358 4375


## SUMMARY


Full stack engineer with 4+ years of experience, working mainly in TypeScript, Node.js and React. At Linfox, migrated a logistics platform's user and permission data from Microsoft Dataverse to PostgreSQL on AWS RDS, cutting authentication time by over 50% for 3,000+ users. Work across serverless backends (AWS Lambda), database design and CI/CD. Previously built healthcare and e-commerce platforms. Melbourne-based Australian permanent resident.


## FUNCTIONAL EXPERTISE

- **TypeScript, JavaScript, Node.js, React**
- **AWS Lambda, RDS, Serverless Framework**
- **PostgreSQL, Prisma, Snowflake**
- **CI/CD, GitHub Actions, Docker**
- **Integration Testing, REST APIs**
- **Performance Engineering (database, build size)**

## FUNCTIONAL EXPERTISE DEMONSTRATED

Linfox

Feb 2025 – Present

*Australia's largest privately owned logistics company, providing supply chain solutions to major retailers, manufacturers, and governments.*

**Full Stack Developer**

Works on modernising internal logistics platforms with a serverless architecture, enhancing usability, performance, modularity.

- Took ownership of backend data layer design by implementing PostgreSQL + Prisma within AWS Lambda serverless architecture, building a scalable and type-safe system supporting 3,000+ users in a logistics platform.
- **Performance optimization**: Migrated critical user-contract data from Microsoft Dataverse to PostgreSQL, deploying the new database to AWS RDS and integrating it directly with Lambda functions. This cut average authentication response time by over 50%. Resolved deployment bottlenecks by replacing legacy bundling with ESbuild, reducing artifact size from 100MB+ to under 20MB and enabling faster, more reliable production releases.
- **Developer experience:** Built and standardised a local Serverless development environment, significantly improving debugging efficiency and reducing reliance on cloud deployments within a 2–3 person engineering team.
- **CI/CD pipeline**: Designed and integrated end-to-end integration testing framework into CI/CD pipelines, reducing regression risk and improving overall system stability.
- **Database upgrades:** Completed a major Prisma ORM version upgrade, managing breaking changes while keeping production stable.
- **Data warehousing:** Managed and optimised the Snowflake data warehouse so operational reporting reads data efficiently.
- **Frontend performance:** Migrated Material UI table views to AG Grid, adding sorting and column reordering; used TanStack Query caching to cut redundant requests during navigation, and Zod to validate request payloads.
- Led end-to-end migration of 10,000+ records (users, permissions, customer data) with zero data loss, ensuring seamless transition and uninterrupted business operations.
- Cross-functional coordination: Contributed to system design decisions and worked closely with cross-functional team members to prioritise and deliver scalable, production-ready solutions.

AUXITA

Jan 2023 – Nov 2024

*Healthcare technology company founded in 2015, partnering with MCK, Pfizer, and Innomar to develop innovative digital health solutions.*

**Software Engineer**

Develops platform features, implements integrations, and enhances system usability for healthcare providers.

- **Problem-solving:** Engineered innovative solutions to expand platform accessibility, integrating Azure AI tools to automate PDF enrollment and onboard 10,000+ potential users.
- **Cross-functional collaboration:** Developed Prescriber admin portal and fax enrollment features, enabling seamless integration of external prescribers and reducing administrative workload by 80%.
- **Programming in MERN stack:** Implemented platform features using multiple programming languages, creating scalable solutions for prescriber engagement and system efficiency.

TECKS
Dec 2021 – Jan 2023
*A compact IT service provider with 5 employees delivering technology solutions for digital businesses.*
**Full Stack Developer**
Developed and optimized e-commerce platform for Future Golf, enhancing website performance and user experience.

- **Programming in PHP**: Engineered new golf tee time booking page using PHP, reducing average booking time by over 50% through optimized design and functionality.
- **Proficiency in SQL**: Integrated SendGrid email APIs and templates, streamlining email communication processes and reducing template modification time by 40%.
- **Cross-functional Collaboration**: Worked closely with a team of 2 developers to implement website enhancements and resolve technical challenges for Future Golf platform.

## PROJECTS

Shopify × YunExpress Cross-border Order Automation · TypeScript, Node.js, SQLite, Express.js, GitHub Actions · 2025

- Built and deployed a production automation system for a cross-border e-commerce business, integrating multiple third-party APIs to eliminate manual order fulfilment workflows.
- Multi-API integration: Orchestrated three external systems with distinct authentication mechanisms — Shopify Admin REST API (OAuth2 client credentials), YunExpress OpenAPI (HMAC-signed requests), and YunExpress OMS internal API (reverse-engineered via DevTools network capture) — enabling fully automated order forwarding with tracking number assignment and Shopify fulfilment.
- Reliability engineering: Implemented idempotent order processing with SQLite state machine, cursor-based incremental sync, and automatic retry logic, ensuring zero duplicate shipments and self-healing on transient failures.
- Operational tooling: Built a password-protected management dashboard for non-technical business partners, featuring order forwarding, packing slip printing, pickup scheduling, and daily WeChat push notifications via scheduled cron jobs (Server酱).
- Production deployment & CI/CD: Self-hosted on DigitalOcean VPS with PM2 process management, Caddy reverse proxy with Let's Encrypt SSL, and GitHub Actions pipeline for automated deployment on every push to main.

## EDUCATION

**Master of Science (Information Technology and Systems).** University of Tasmania (Tas, Australia)

**Bachelor of Arts (Russian Language and Literature).** Beijing International Studies University (Beijing, China)

## OTHER RELEVANT INFORMATION

**Additional Languages:** Mandarin (native), Russian (fluent)
**Tools**: Git, Agile methodology, Jira